

UNIT 7**SECURITY IN CLOUD COMPUTING****7.1. Cloud Computing Challenges:**

Cloud computing, an emergent technology, has placed many challenges in different aspects of data and information handling. Some of these are shown in the following diagram:

**Security and Privacy**

Security and Privacy of information is the biggest challenge to cloud computing. Security and privacy issues can be overcome by employing encryption, security hardware and security applications.

Portability

This is another challenge to cloud computing those applications should easily be migrated from one cloud provider to another. There must not be vendor lock-in. However, it is not yet made possible because each of the cloud provider uses different standard languages for their platforms.

Interoperability

It means the application on one platform should be able to incorporate services from the other platforms. It is made possible via web services, but developing such web services is very complex.

Computing Performance

Data intensive applications on cloud requires high network bandwidth, which results in high cost. Low bandwidth does not meet the desired computing performance of cloud application.

Reliability and Availability

It is necessary for cloud systems to be reliable and robust because most of the businesses are now becoming dependent on services provided by third-party.

7.2. Cloud Security Challenges:**1. Data Related Security Data Breach: 1. Confidentiality 2. Integrity**

Data Lock in: Users may lose data if they migrate from one vendor to another vendor.
Data Remanence: It is the residual representation of data that have been nominally erased or removed in some way.

2. Data Related Security Data Recovery:

Sometimes server may break down and cause damage or loss to users' data. To avoid this, data should be backed up to be recovered in future

Data Locality: In SaaS model of cloud environment, the user doesn't know where the data is stored which may be an issue. The issue can be solved by creating secure SaaS model which can provide reliability to the customer on the location of the data of the user.

3. Application related security issues

Cloud malware injection attack: In this attack a malicious virtual machine or a service implementation is injected into the cloud system. one solution to prevent this is to perform the integrity check to the service instance. **Cookie poisoning:** In this an unauthorized access is made into the application by modifying the contents of the cookie. One solution is to clean up the cookie or encrypt the cookie data.

Backdoor and Debug Option: Debug option is for the developers who use it to implement any changes requested at later stage in a website since these debug option provides back entry for the developers, sometimes these debug options are left enabled unnoticed, they may provide easy access to the hackers and allow them to make changes in the website. **Hidden Field Manipulation:** Certain fields are hidden in the web-site and is used by the developers. Hacker can easily modify on the web page.

CSP level attacks Guest hopping attack: An attacker will try get access to one virtual machine by penetrating another virtual machine hosted in the same hardware. **SQL injection:** It can be done by injecting the SQL commands into the database of an application to crash the database.

CSP level attacks Malicious Insider: In private cloud, its employee is granted access to the sensitive data of some or all customer administrators. Such privileges may expose information to security threats. **Side channel attack:** It occurs when an attacker places a malicious virtual machine on the same physical machine as the victim machine so that he can access all the confidential information on the victims machine.

4. Network level attacks:

DNS attacks: Domain hijacking: Domain hijacking is defined as changing the name of a domain without the knowledge or permission from the domain's owner or creator. This enables the intruders to access the sensitive information. Cross site scripting: It is a type of attack in which user enters right URL of a website and hacker on the other site redirect the user to its own website and hack its credentials.

IP spoofing's attack: When hackers overflow a network server or web server with frequent request of services to damage the network, the denial of service cannot keep up with them, server could not legitimate client regular requests.

Man in the middle attack: This is another issue of network security that will happen if secure socket layer (SSL) is not properly configured. Network Sniffing: Another type of attack is network sniffer, it is a more critical issue of network security in which unencrypted data are hacked through network.

7.3. Cloud Security Monitoring:

As the name already suggests, Cloud security monitoring is the process of protecting the data stored in the cloud from any unauthorized access.

When it comes to cloud security and management, monitoring the cloud infrastructure is one of the most critical components.

The whole concept of cloud security management revolves around the experts assessing the security tools and modules the organization has implemented. Further, helping the company to enhance their strength while reducing threats to their IT infrastructure.

Cloud monitoring includes both automated as well as manual assessment, which involves monitoring websites, servers, software platforms, and mobile apps. Moreover, for proper cloud security management, the solution has to reach all aspects of the cloud, including the cloud data sharing applications, cloud access management tools, and others.

Needs for Cloud Security Monitoring:

The criticality of monitoring in cloud security and management has grown from being a caution or threat detection to an essential aspect. This has increased the reliability of cloud service providers on automated solutions for real-time assessment and measurement of data, applications, or infrastructure behaviours to identify potential security threats.

1. Keeping Data Secure:

Cloud security monitoring has increasingly facilitated the identification of patterns and high-risk vulnerabilities in an infrastructure. Additionally, the reluctance of companies to the adoption of cloud computing on the back of control-related uncertainties has been ebbing away, with the promise of effective security monitoring to make the data transfer and storage more convenient.

2. Continuous Monitoring & Securing:

Cloud monitoring helps to review online computing resources for workflow optimization while ensuring ceaseless service availability and high performance of servers, websites, and application security monitoring. It also aids in the evaluation of resource availability, program speed, and resource levels, to further help predict vulnerability attacks before their occurrence.

3. Increased Benefits

Companies adopting cloud monitoring have found a perfect balance between mitigating risks and leveraging the benefits of the cloud environments, without any hindrances to business processes.



Key Best Practices for Effective Cloud Security Monitoring

- ✓ Stringent Data Control
- ✓ Securing the Code
- ✓ Separating Crucial Metrics
- ✓ Patch Management
- ✓ Automation
- ✓ Identity and Access Management
- ✓ Data Protection

1. Stringent Data Control

One of the most efficacious measures for mitigating cloud security risks is gaining complete control over data across all endpoints. Using solutions that scan, assess and take action on the data sources, before its transition from an enterprise network, enables a robust defense against any data loss through the cloud provider.

This further helps in avoiding attack surface vulnerabilities, such as uploading of sensitive files to unprotected cloud repositories.

2. Securing the Code

Securing codes must be one of the top priorities, to eliminate risks from potential cyberthreats. During the development of codes for websites, the focus lies in selecting the right security development lifecycle (SDL), which aligns well with the delivery strategies of companies. Key benefits associated with secure SDL include continuous security, early risk detection, threat detection, improved stakeholder awareness, and reduction of business risks.

3. Separating Crucial Metrics

A software developer aims at learning features of cloud security monitoring while designing new software with specific security aspects. For the implementation of a monitoring strategy, key considerations must be emphasized such as

- Determination of the resource inventory utilized in the organization
- Mapping all attributes of data that the organization aims to gather
- Mutual decision appertaining to software that best suits the organization

While several cloud security monitoring solutions have been launched, the above guidelines would help in the successful execution of online monitoring system practices.

4. Patch Management

Unpatched software and system result in significant issues. Keeping an organization's environment secure by updating the systems regularly has become imperative.

Key requirements for an organization to eliminate risks for their systems include asset inventory, vulnerability data gathered, and configuration for understanding the criticality of the risks associated. Organizations must consider preparing the checklist of important procedures and make sure to keep all updates in check for arresting the vulnerabilities before implementation into the live environment. An apt blend of vulnerability scanning and automatic patching allude lower impact of threats.

5. Automation

Scripting remains an essential feature in the cloud security monitoring strategy, as monitoring and reporting processes can be scripted for enabling automation of systems. Organizations must focus on the implementation of the monitoring software owing to the virtual nature of operations on the cloud environments, even as automating the red-flag and logging system for real-time alerts. In simple terms, Cloud automation is the activity or discipline of automating part or all of the manual operations associated with managing and operating cloud-based IT infrastructure using specific software tools and methodologies.

6. Identity and Access Management

Identity and access management provides capabilities of provisioning and access control for users. Administrators can use the IAM to manage users and groups and implement granular permission rules to limit access to resources and APIs.

Organizations must ensure the policies created are attached to roles or groups instead of individual users to deprive unnecessary or excessive privileges or permissions to users. To stem any unauthorized access to resources, organizations must provision access to these resources using roles instead of giving separate set of credentials for access.

It must be made sure that users are provided minimal access privileges to resources, which would not restrict them to fulfil their responsibilities. All users must have a multifactor authentication for their individual accounts as a defensive strategy against any compromised account.

Organizations must also limit the number of users with administrative privileges. The access keys must be rotated on a regular basis, and standardized periodically for password expiration, ensuring inaccessibility to data sources with a potential stolen or lost key.

A strong password policy must be enforced, requiring a minimum of fourteen characters that contain at least one symbol, one uppercase letter and one number. Also, a password reset policy must be enforced to prevent users from using the password used in the past 24 resets.

7. Data Protection

New technologies entail distinct issues and with the cloud-based storage solutions on the fore, data protection has become an imperative challenge to be addressed.

Organizations must opt for the cloud service providers that deliver data encryption as a standard feature in their offerings. Encrypting highly sensitive data including personally identifiable information (PII) or protected health information with the aid of customer-controlled keys has become a must-have for organizations eyeing the move to cloud computing. Customer-controlled key put the pressure of management on the customer, however they provide better control. Organizations must choose the provider who guarantees protection against loss of critical data and meet the needs of data backup and recovery. Data replication is a common practice among the cloud service providers to ensure persistence. Organizations must thoroughly analyze the cloud deployment to identify where the replicated data has been cached or stored and take actions accordingly to ensure that the copied data has been deleted.

7.4. Cloud Risk Management pillars

1. Cybersecurity

As the complexity and frequency of cyber threats rise, organizations should create a comprehensive cybersecurity program. They should concentrate on finding vulnerabilities, deploying solutions to protect important business data, detecting potential threats that have infiltrated the infrastructure, and assisting essential business applications and systems in responding to and recovering from incidents.

2. Regulatory compliance

In light of problems, authorities all over the world are continuing to act by releasing and updating recommendations on cloud computing and how to avoid and respond to cyber-attacks. Without automation, the expenses of maintaining a risk staff to remain on top of these rules will skyrocket.

3. Backup and recovery

Almost every company does frequent backups. However, very few businesses actually undertake frequent restoration to ensure the functionality and sufficiency of backups, resulting in unpleasant shocks at the last minute. Cloud companies have this step-down path since the consequences of a blunder will be devastating to their business. Again, this is a two-edged sword that is depending on the cloud provider's rules, which may or may not be sufficient for your organization's needs.

4. Platform support

Many companies are unable to roll out patches on time, or even discover the appropriate patches, for a variety of reasons such as a lack of a suitable knowledge base, time, or testing infrastructure. Most cloud providers do not have these weaknesses, guaranteeing that the platforms and apps you use on such cloud settings are properly up to date. This is a two-edged sword because vulnerabilities are found in several cloud providers. Organizations with reasonably developed procedures ensure things such as timely internal system changes and sufficient testing. The same cannot be true for cloud providers owing to a lack of visibility and openness.

5. Vendor Management

The inclusion of third-party suppliers in cloud business models has raised security issues. Many cloud providers are undergoing official third-party security assessments, such as the International Organization for Standardization (ISO), Service Organization Control (SOC) 2, and the Federal Risk Authorization and Management Program (FedRAMP). To prevent security problems, you should concentrate on establishing a corporate public cloud strategy that includes security guidelines on approved SaaS usage. It is important to employ a life-cycle governance model that stresses ongoing operational management of your public cloud utilization.

6. Cloud Migration

The process of transferring apps, data or even the whole corporate IT infrastructure to distant server facilities and a virtual environment is known as cloud migration. The benefits of cloud migration are numerous. The cloud architecture allows for the acceptance of any workload, and the simplicity with which new services may be added allows for rapid response to changing business demands. Several companies are still delaying cloud adoption due to dependability and security concerns. Otherwise, people might not find it useful. Organizations can save money by migrating to the cloud. Companies may save a lot of money by migrating to the cloud, especially in the long run. When compared to on-premise hardware, the cloud requires no initial expenditure. When employing on-premise infrastructure, you may encounter capacity difficulties. However, by utilizing cloud technology, you may entirely eliminate capacity issues.

7.5. Cloud computing—key legal issues

Following are the legal and commercial issues relating to cloud services:

- Due diligence
- Contract documents
- Service description
- Service levels
- Data protection
- Data portability and backup
- Security
- Record keeping and audit rights
- Compliance with laws
- Intellectual property
- Supplier liability for content
- Liability under the contract
- Variation of terms
- Suspension and termination rights
- Business continuity and disaster recovery
- Export control laws
- Sector specific regulation and guidance
- International laws

7.6. Cloud Incident Response Plan:

It is simply the process used to manage cyber-attacks in a cloud environment. There are several key aspects of a cloud incident response system that differentiate it from a non-cloud incident response system, notably in the areas of governance, shared responsibility, and visibility

Lifecycle of Cloud incident Response Plan:

- Preparation
- Detection and analysis
- Containment, eradication, and recovery
- Post-mortem

Creating an incident response plan

- An incident response plan should be set up to address a suspected data breach in a series of phases. Within each phase, there are specific areas of need that should be considered.

The incident response phases are:

Preparation
Identification
Containment
Eradication
Recovery
Lessons Learned

1. Preparation

This phase will be the work horse of your incident response planning, and in the end, the most crucial phase to protect your business.

Part of this phase includes:

Ensure your employees are properly trained regarding their incident response roles and responsibilities in the event of data breach. Develop incident response drill scenarios and regularly conduct mock data breaches to evaluate your incident response plan. Ensure that all aspects of your incident response plan (training, execution, hardware and software resources, etc.) are approved and funded in advance.

2. Identification

This is the process where you determine whether you've been breached. A breach, or incident, could originate from many different areas. The question addressed should be :

When did the event happen?

How was it discovered?

Who discovered it?

Have any other areas been impacted?

What is the scope of the compromise?

Does it affect operations?

Has the source (point of entry) of the event been discovered?

3. Containment

When a breach is first discovered, your initial instinct may be to securely delete everything so you can just get rid of it. However, that will likely hurt you in the long run since you'll be destroying valuable evidence that you need to determine where the breach started and devise a plan to prevent it from happening again. Instead, contain the breach so it doesn't spread and cause further damage to your business. If you can, disconnect affected devices from the Internet. Have short-term and long-term containment strategies ready. It's also good to have a redundant system back-up to help restore business operations. That way, any compromised data isn't lost forever. Change all user and administrative access credentials and harden all passwords.

4. Eradication

Once you've contained the issue, you need to find and eliminate the root cause of the breach. This means all malware should be securely removed, systems should again be hardened and patched, and updates should be applied. Whether you do this yourself, or hire a third party to do it, you need to be thorough. If any trace of malware or security issues remain in your systems, you may still be losing valuable data, and your liability could increase.

5. Recovery

This is the process of restoring and returning affected systems and devices back into your business environment. During this time, it's important to get your systems and business operations up and running again without the fear of another breach.

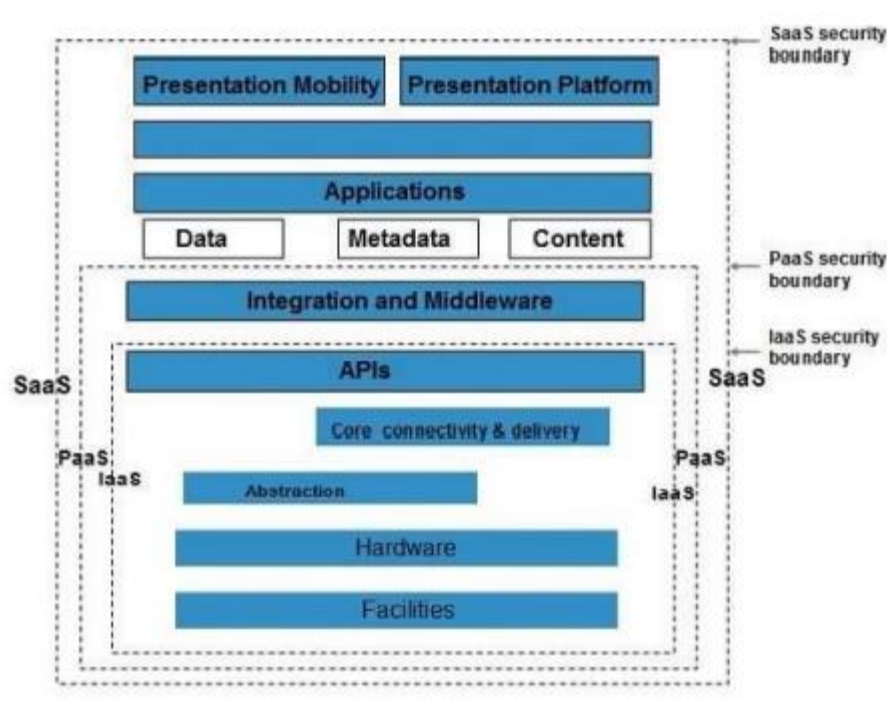
6. Lessons Learned

Once the investigation is complete, hold an after-action meeting with all Incident Response Team members and discuss what you've learned from the data breach. This is where you will analyze and document everything about the breach. Determine what worked well in your response plan, and where there were some holes. Lessons learned from both mock and real events will help strengthen your systems against the future attacks.

7.7. Understanding Security of Cloud Security Boundaries:

A particular service model defines the boundary between the responsibilities of service provider and customer. Cloud Security Alliance (CSA) stack model defines the boundaries between each service model and shows how different functional units relate to each other.

The following diagram shows the CSA stack model:



Key Points to Cloud Security Architecture Model

- IaaS is the most basic level of service with PaaS and SaaS next two above levels of services.
- Moving upwards, each of the service inherits capabilities and security concerns of the model beneath.
- IaaS provides the infrastructure, PaaS provides platform development environment, and SaaS provides operating environment.
- IaaS has the least level of integrated functionalities and integrated security while SaaS has the most.
- This model describes the security boundaries at which cloud service provider's responsibilities end and the customer's responsibilities begin.

- Any security mechanism below the security boundary must be built into the system and should be maintained by the customer.
- Although each service model has security mechanism, the security needs also depend upon where these services are located, in private, public, hybrid or community cloud.

7.8. Vulnerability Assessment:

A vulnerability assessment is a systematic review of security weaknesses in an information system. It evaluates if the system is susceptible to any known vulnerabilities, assigns severity levels to those vulnerabilities, and recommends remediation or mitigation, if and whenever needed.

Examples of threats that can be prevented by vulnerability assessment include:

- ❖ SQL injection, XSS and other code injection attacks.
- ❖ Escalation of privileges due to faulty authentication mechanisms.
- ❖ Insecure defaults software that ships with insecure settings, such as a guessable admin password.

There are several types of vulnerability assessments. These include:

- ✓ **Host assessment** – The assessment of critical servers, which may be vulnerable to attacks if not adequately tested or not generated from a tested machine image.
- ✓ **Network and wireless assessment** – The assessment of policies and practices to prevent unauthorized access to private or public networks and network-accessible resources.
- ✓ **Database assessment** – The assessment of databases or big data systems for vulnerabilities and misconfigurations, identifying rogue databases or insecure dev/test environments, and classifying sensitive data across an organization's infrastructure.
- ✓ **Application scans** – The identifying of security vulnerabilities in web applications and their source code by automated scans on the front-end or static/dynamic analysis of source code.

Vulnerability assessment: Security scanning process

The security scanning process consists of four steps: testing, analysis, assessment and remediation.



1. Vulnerability identification (testing)

The objective of this step is to draft a comprehensive list of an application's vulnerabilities. Security analysts test the security health of applications, servers or other systems by scanning them with automated tools, or testing and evaluating them manually. Analysts also rely on vulnerability databases, vendor vulnerability announcements, asset management systems and threat intelligence feeds to identify security weaknesses.

2. Vulnerability analysis

The objective of this step is to identify the source and root cause of the vulnerabilities identified in step one. It involves the identification of system components responsible for each vulnerability, and the root cause of the vulnerability. For example, the root cause of a vulnerability could be an old version of an open-source library. This provides a clear path for remediation – upgrading the library.

3. Risk assessment

The objective of this step is the prioritizing of vulnerabilities. It involves security analysts assigning a rank or severity score to each vulnerability, based on such factors as:

Which systems are affected?

What data is at risk?

Which business functions are at risk?

Ease of attack or compromise.

Severity of an attack.

Potential damage as a result of the vulnerability.

4. Remediation

The objective of this step is the closing of security gaps. It's typically a joint effort by security staff, development and operations teams, who determine the most effective path for remediation or mitigation of each vulnerability.

Specific remediation steps might include:

Introduction of new security procedures, measures or tools.

The updating of operational or configuration changes.

Development and implementation of a vulnerability patch.

7.9. Data security:

- ✓ It refers to the process of protecting data from unauthorized access and data corruption throughout its lifecycle.

Data security includes data encryption, hashing, tokenization, and key management practices that protect data across all applications and platforms.

Why Data Security?

- ✓ Organizations around the globe are investing heavily in information technology (IT) cyber security capabilities to protect their critical assets. Whether an enterprise needs to protect a brand, intellectual capital, and customer information or provide controls for critical infrastructure, the means for incident detection and response to protecting organizational interests have three common elements: people, processes, and technology.

Data Security Solutions:

They are listed below:

1. **Cloud data security** – Protection platform that allows you to move to the cloud securely while protecting data in cloud applications.
2. **Data encryption** – Data-centric and tokenization security solutions that protect data across enterprise, cloud, mobile and big data environments.
3. **Hardware security module** -- Hardware security module that guards financial data and meets industry security and compliance requirements.
4. **Key management** -- Solution that protects data and enables industry regulation compliance.
5. **Enterprise Data Protection** – Solution that provides an end-to-end data-centric approach to enterprise data protection.
6. **Payments Security** Solution provides complete point-to-point encryption and tokenization for retail payment transactions, enabling PCI scope reduction.
7. **Mobile App Security** - Protecting sensitive data in native mobile apps while safeguarding the data end-to-end.
8. **Web Browser Security** - Protects sensitive data captured at the browser, from the point the customer enters cardholder or personal data, and keeps it protected through the ecosystem to the trusted host destination.
9. **Email Security** – Solution that provides end-to-end encryption for email and mobile messaging, keeping Personally Identifiable Information and Personal Health Information secure and private.

7.10. Cloud Application Security:

❖ Cloud application security is a series of defined policies, processes, controls, and technology governing all information exchanges that happen in collaborative cloud environments like Microsoft Office 365, Google G Suite, Slack, etc

Cloud application security threats

They are

1. Misconfiguration of application setup is the single biggest threat to cloud security because data breaches tend to happen when services are accidentally exposed to the public internet.
2. Unauthorized access to a website, server, service, or other system is also an area for great concern because once they're in, there's no telling what unauthorized users will do to create chaos.
3. Insecure APIs and interfaces present easy opportunities for attackers to breach systems because they are the only asset(s) outside of the organizational boundary with a public IP address.
4. Account hijacking is feared because so much sensitive data and resources is stored and accessed on devices shared by many different users—and because keeping tabs on rogue employees is difficult.

7.11. VM security:

To secure your virtual machines, keep the guest operating systems patched and protect your environment just as you protect your physical machine. Consider disabling unnecessary functionality, minimize the use of the virtual machine console, and follow other best practices.

1. **Protect the guest operating system**

To protect your guest operating system, make sure that it uses the most recent patches and, if appropriate, anti-spyware and anti-malware applications. See the documentation from your guest operating system vendor and, potentially, other information available in books or on the Internet for that operating system.

2. **Disable unnecessary functionality**

Check that unnecessary functionality is disabled to minimize potential points of attack. Many of the features that are used infrequently are disabled by default. Remove unnecessary hardware and disable certain features such as host-guest filesystem (HGFS) or copy and paste between the virtual machine and a remote console.

3. **Use templates and scripted management**

Virtual machine templates enable you to set up the operating system so that it meets your requirements, and to create other VMs with the same settings. If you want to change virtual machine settings after initial deployment, consider using scripts, for example, PowerCLI. This documentation explains how to perform tasks using the GUI. Consider using scripts instead of the GUI to keep your environment consistent.

4. **Minimize use of the virtual machine console**

The virtual machine console provides the same function for a virtual machine that a monitor on a physical server provides. Users with access to a virtual machine console have access to virtual machine power management and to removable device connectivity controls. As a result, virtual machine console access might allow a malicious attack on a virtual machine.

5. **Consider UEFI secure boot**

You can configure your virtual machine to use UEFI boot. If the operating system supports secure UEFI boot, you can select that option for your VMs for additional security.

6. **Consider Carbon Black Cloud Workload**

You can install and use Carbon Black Cloud Workload to identify risk, prevent attacks, and detect unusual activities. With the AppDefense functionality built in to the Carbon Black Cloud platform, Carbon Black Cloud Workload is the successor product for AppDefense.

7.12. Cloud disaster recovery:

- Cloud disaster recovery (CDR) is a cloud-based managed service that helps you quickly recover your organization's critical systems after a disaster and provides you remote access to your systems in a secure virtual environment.

- When it comes to traditional DR, managing a secondary data center can be time consuming and costly.
- Cloud disaster recovery has changed everything when it comes to traditional DR by eliminating the need for traditional infrastructure and significantly reducing downtime.
- IT departments can now harness the power of the cloud for immediate spin-up and fail over. This creates faster recovery times at a fraction of the cost.

Working of CDR

- Cloud disaster recovery takes a very different approach than traditional DR.
- Instead of loading the servers with the OS and application software and patching to the last configuration used in production, cloud disaster recovery encapsulates the entire server, which includes the operating system, applications, patches, and data into a single software bundle or virtual server.
- The virtual server is then copied or backed up to an offsite data center or spun up on a virtual host in minutes.
- Since the virtual server is not dependent on hardware, the operating system, applications, patches, and data can be migrated from one data center to another much faster than traditional DR approaches.

Types of Disaster

There are three main categories of disaster that can affect businesses:

- **Natural disasters:** Natural disasters such as floods or earthquakes are rarer but not infrequent. If a disaster strikes an area that contains a server that hosts the cloud service you're using, this could disrupt services and require disaster recovery operations.
- **Technical disasters:** Perhaps the most obvious of the three, technical disasters encompass anything that could go wrong with the cloud technology. This could include power failures or a loss of network connectivity.
- **Human disasters:** Human failures are a common occurrence and are usually accidents that happen whilst using the cloud services. These could include inadvertent misconfiguration or even malicious third-party access to the cloud service.

DR working and methodologies:

Disaster recovery in cloud computing is a delicate process.

The methodologies behind them must be understood carefully for successful recovery.

Backup and restore

Backing up data and restoring it is one of the easiest, cheapest and fastest ways to recover from a cloud computing disaster. This can be mainly used to mitigate regional disasters such as natural disasters by replicating the data and storing it in a geographically different location.

Pilot Light

The 'Pilot Light' disaster recovery approach is a method where your company replicates only the minimal and core services it needs to function. This means that only a small part of your IT structure needs to be replicated and provides a minimally functional replacement in case of disaster

Warm Standby

The warm standby approach is when a scaled down version of your fully functional environment is available and always running in a separate location to your main server. This means that in the event of a disaster, your company can still run a version of the site that is based in a different region.

Multi-site deployment

Although the most expensive solution of the three, multi-site deployment provides the most comprehensive solution to regional disasters. Multi-site deployment involves running your full workload simultaneously in multiple regions. These regions can be actively used or on a standby in case of disaster in a different region.

Benefits of cloud computing disaster recovery in the cloud:

- Cloud-based disaster recovery is much faster than on-premises disaster recovery and doesn't require as much complexity. This simplicity also allows for **easy testing of the disaster recovery services**, so your company can make sure your disaster recovery plans are fully functional.
- The presence of cloud providers also reduces the workload from your company as **the operational burden is essentially outsourced**.
- Cloud-based services also offer opportunities to automate, reducing human error and improving service recovery times.
- One of the greatest benefits of cloud-based disaster recovery is the option to mix-and-match recovery options. Choosing a mixture of methodologies based on RTO and RPO allows you to minimize costs whilst being able to use all the services you need

DR recovery plan:

Here are 5 steps that can help you prepare a recovery plan:

1. Your disaster recovery plan should be part of your business continuity plan.

This should involve definitions of RTO and RPO to help you decide which cloud services you'll need and improve cost efficiency.

2. If you haven't done so already, define the RTO and RPO for your disaster recovery.

This forms the basis of your disaster recovery plan and, in turn, the kinds of disaster recovery services you'll need.

3. Design your plan with your recovery goals in mind.

This involves looking at your RTO and RPO points to decide which disaster recovery pattern you'll need to meet those criteria. Your recovery goals should outline the maximum and minimum affects to your services

4. Design for end-to-end recovery.

Your plan should include recovery for every aspect of your business that needs to be operational.

5. Create specific tasks to ensure a smooth-running process.

The more specific your tasks are, the easier the recovery process will be and the fewer chances there will be of deviating from the plan. Developing and implementing best practices for cloud disaster recovery are key to a successful operation.